VIT
Vellore Institute of Technology
(Deemed to be University under section 3 of UGC Act, 1956)
CHENNAI

Reg. Number:	

Continuous Assessment Test (CAT) – I AUGUST 2025

Programme	:	B.Tech BAI/BDS/BLC/BPS/BRS	Semester	:	Fall Semester 2025 – 2026
Course Code & Course Title	:	BCSE204L Design and Analysis of Algorithms	Class Number	:	CH2025260101320,CH2025260101448 CH2025260101449,CH2025260101593 CH2025260101595,CH2025260101598 CH2025260102084,CH2025260102085 CH2025260102086,CH2025260102087 CH2025260102088
Faculty	:	Dr. Aarthi B, Dr. Jannath Nisha O S Dr. Ranjith Kumar M, Dr. Raja M, Dr. Indira B, Dr. Lekshmi K Dr. Revathi A R, Dr. Sivaranjani N Dr. Helen Vijitha P, Dr. Sarita Kumari Dr. Kaja Mohideen A	Slot	:	G2+TG2
Duration	:	90 Minutes	Max. Mark		50

General Instructions:

- Write only your registration number on the question paper in the box provided and do not write other information
- Use statistical tables supplied from the exam cell as necessary
- Use graph sheets supplied from the exam cell as necessary
- Only non-programmable calculator without storage is permitted
- If any assumptions are required, assume the same and mention those assumptions in the answer script.
- The design component should consist of logic to develop and illustrate the pseudocode.
- The analysis component should consist of the computation of T(n) and time complexity.

Answer all questions

Q. No	Sub Sec.	Description	Marks	CO	BT Level
1		A marine biologist is studying the daily depth readings (in meters) of a tagged dolphin, stored in an array D[1:n]. The biologist wants to find the most dramatic 3-day diving pattern where: $i < j < k$. The difference D[j] - D[i] is negative (dolphin dives deeper). The difference D[k] - D[j] is positive and at least 5 meters (dolphin resurfaces sharply). For each valid triplet (i, j, k) meeting this condition, the dive score is calculated as: $\|D[j] - D[i]\| + \|D[k] - D[j]\|$. The task is to design a brute force pseudocode to find one valid triplet with the maximum dive score. If no such triplet exists, output "No valid triplet". For your illustration consider the input n = 6, D = [20, 25, 15, 10, 17, 30]. [Rubrics: Logic: 2 marks, Illustration: 2 marks, Pseudocode: 4 marks, Time-complexity: 2 marks]	10	CO1	K3
2		A common subsequence Z of two strings P and Q is said to be a Unique Common Subsequence (UCS) if no character repeats in Z. Given two strings P of length n and Q of length m, design a dynamic programming pseudocode to compute all the UCS of P and Q whose length is greater than (m + n) / 3 (use the floor	10	CO2	K3

		function). If no UCS exists with the specified length, the pseudocode should return −1. Your design components should contain all the required steps and derive the time complexity of the algorithm. For illustration consider the given inputs P = ABCDFGHIEJKL, Q = XAYBCGHDZIEKL [Rubrics: Logic: 2 marks, Illustration: 2 marks, Pseudocode: 4 marks, Time-complexity: 2 marks]			
3		An inversion in an array A[ ] of length n is defined as a pair of indices (i,j) such that i < j and A[i] > A[j]. In simple terms, an inversion occurs when a larger number appears before a smaller number in the array. For example, in the array A = [40, 30], there is one inversion because 40 > 30 and index(40) < index(30), whereas in A= [30, 40] there is no inversion because 30 < 40. Given an integer n and an array A of n integers, design a pseudocode using the Divide and Conque technique to count the total number of inversions in A. Your design technique should clearly include all the required steps to find the total inversion count in A. For illustration, consider the case where n = 6 and A = [50, 40, 60, 10, 20, 30], and trace your algorithm to show how inversions are counted. Analyse your algorithm with respect to its time complexity. [Rubrics: Logic: 3 marks, Illustration: 4 marks, Pseudocode: 5 marks, Time-complexity: 3 marks]	15	CO2	K3
4		Algorithm ABC(arr[0..n-1]) n ← length(arr) Create an empty array d for i ← 0 to n - 2 do if arr[i] - arr[i + 1] >= 0 then d[i] ← arr[i] - arr[i + 1] else d[i] ← arr[i + 1] - arr[i] for i ← 0 to length(d) - 1 do for j ← i + 1 to length(d) - 1 do if d[i] < d[j] then temp ← d[i] d[i] ← d[j] d[j] ← temp return d Given Input: arr = [10, 4, 15, 3, 8] (a)Trace the output of the algorithm. (4 marks) (b)Explain the main functionality of the algorithm. (3 marks) (c)Find the step count of the given algorithm. (4 marks) (d)Derive the time complexity of the algorithm using Big-O notation.(4marks)	15	CO1	K2

**********All the best **********